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NeuroTrace

Continuous Physiological & Environmental Headache Tracking

TEAM NOVATRIX

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NeuroTrace — Continuous Physiological & Environmental Headache Tracking

Proposed Solution

- Compact AI-enabled wristband continuously monitors physiological, environmental, and activity data.
- Uses HR/HRV, EDA, skin temperature, motion, and environmental sensors to collect headache-related parameters.
- Stores the user's headache history, duration, activities, and possible triggers to create a personal pattern database.
- Requires only minimum user input instead of daily manual logging.
- Provides early alerts vibration feedback, recommendations and emergency assistance through a button and microphone.

How It Addresses the Problem

- Identifies possible headache triggers such as heat, stress, dehydration, and activity patterns.
- Provides early warnings when sensor conditions match previously identified headache patterns.
- Helps users understand their individual headache causes instead of relying on generalized advice.
- Reduces repeated discomfort by providing personalized preventive suggestions based on collected data.

Innovation & Uniqueness

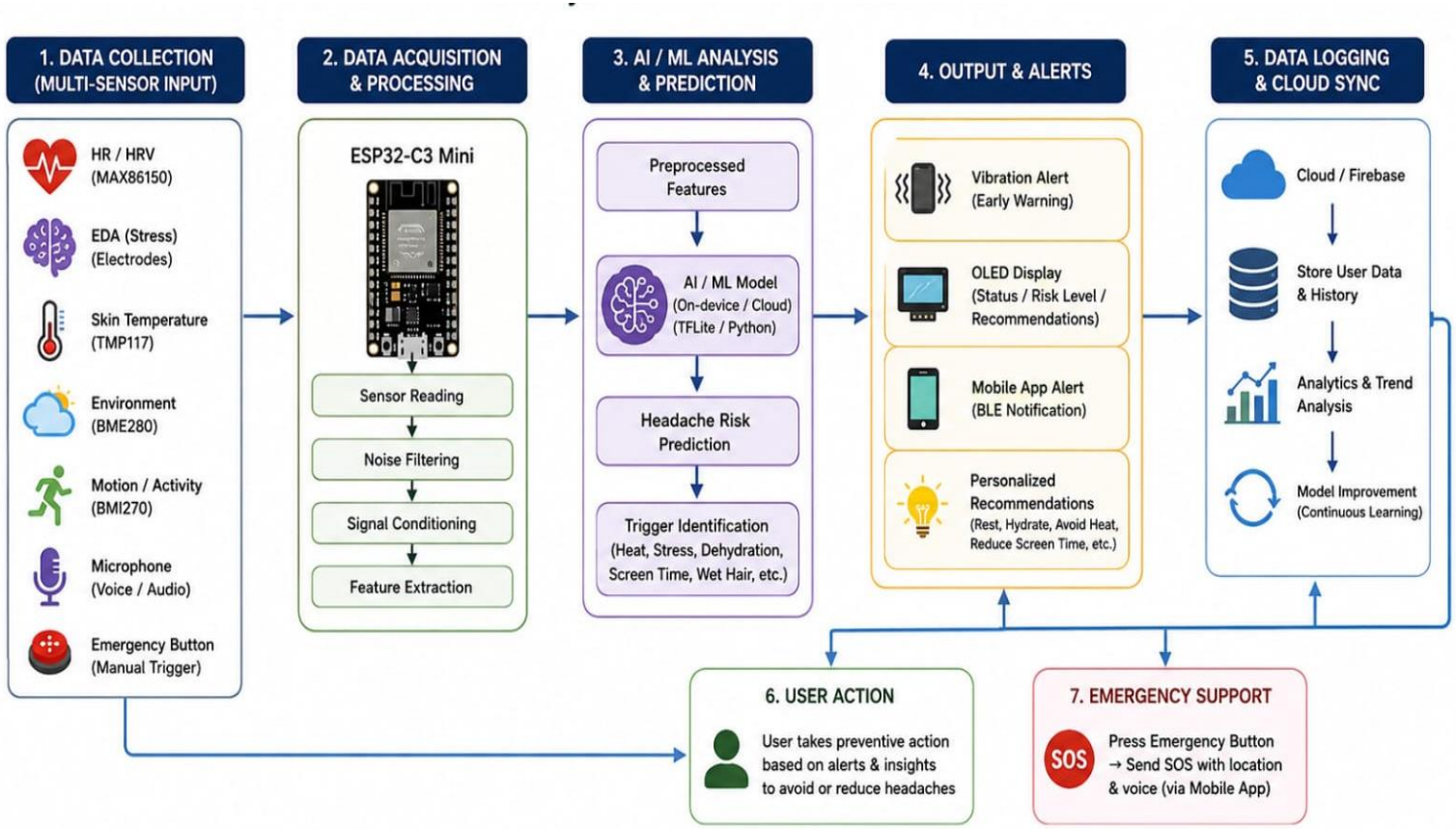
- Combines physiological, environmental and behavioural sensing in a single compact wristband.
- Identifies combinations of multiple factors together(eg.:heat exposure +low hydration)rather than tracking single factors in isolation
- Builds a personalized trigger model in approximately 2 weeks, versus 3+ months for existing dairy-based systems.
- Moves from simple monitoring to predictive, personalized headache prevention.
- Emergency button & microphone work as a dual-purpose safety feature-usable for any risk situation.
- Integrates health monitoring, AI-based prediction, alerts, and emergency communication into one wearable device.

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TECHNICAL APPROACH

ARCHITECTURE

- Sensing: Wearable band collects HR/HRV, motion, temperature & EDA(stress/hydration) data continuously, along with emergency button and microphone input when triggered.
- Processing: ESP32 organizes and timestamps all sensor data, while separately monitoring the emergency button for instant activation..
- Analysis: ML model studies stored historical data to learn personal headache patterns while emergency signals bypass ML for real time response.
- Output: OLED displays preventive suggestions in normal mode; microphone connects the user directly to a caretaker in emergency mode.



COMPONENTS :



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FEASIBILITY AND VIABILITY

ANALYSIS OF THE FEASIBILITY OF THE IDEA

- Technical Feasibility: ESP32 and compact sensors can be integrated into a small wearable band for continuous monitoring.
- AI Feasibility: Personal headache records and sensor data can identify possible triggers and provide early alerts.

POTENTIAL CHALLENGES & RISKS

- Sensor Accuracy: Motion, sweat and poor skin contact may affect sensor readings.
- Battery Limitation: Continuous sensing and Bluetooth communication can reduce battery life.

STRATEGIES TO OVERCOME

- Sensor Calibration: Use proper placement, calibration and signal filtering to improve accuracy
- Power Optimization: Use low-power sensors, sleep modes and adaptive sampling to extend battery life.



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IMPACT AND BENEFITS

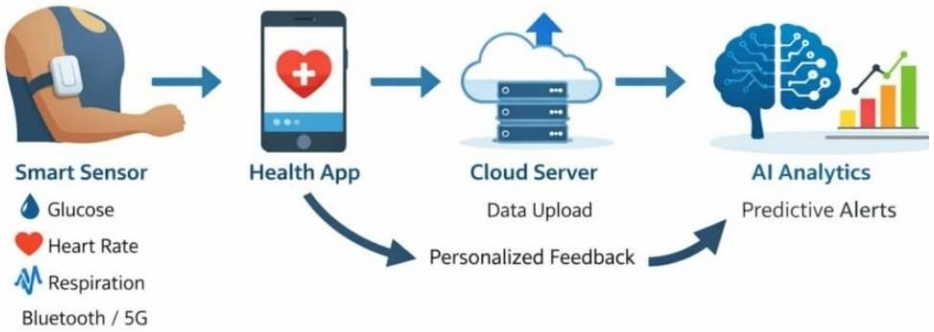
IMPACT TO USER

- Early Warning: Alerts the user before a headache may become severe.
- Personalized Monitoring: Learns individual headache patterns and common triggers.
- Better Daily Awareness: Helps users identify links between stress, heat, activity, and headaches.
- Emergency Support: SOS button and microphone provide quick assistance during critical situations.

BENEFITS OF THE SOLUTION

- Prevention-focused: Helps users take action before symptoms worsen.
- Continuous Monitoring: Tracks multiple health and environmental parameters through a single wristband.
- Personalized Recommendations: Suggests actions based on the user's own recorded patterns.
- Compact & Convenient: Wearable, portable and suitable for everyday use

For Instance:



Example:

When the user is exposed to high heat, the band monitors temperature, HR/HRV and EDA. The AI compares the readings with the user's previous headache patterns and identifies a possible heat-related trigger. The band then gives a vibration alert with a recommendation to hydrate and move to a cooler place.

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RESEARCH AND REFERENCES

Digital Phenotyping for Migraine – 2025 PubMed Link

Wearable Nocturnal Autonomic and Sleep Biomarkers for Predicting Next-Day Headache – 2026 PMC Link

Forecasting Migraine with Time-Series Machine Learning – 2026 PMC Link

AI, Data Science, and Wearable Devices in Headache – 2025 PMC Link

Machine Learning and Wearable Technology – 2024 PMC Link

MiSleepS – Sleep, Stress and Migraine: An Observational and Training Study-2026 PMC Link

Accuracy, Utility and Applicability of the WHOOP Wearable Monitoring Device in Health, Wellness and Performance – 2026 PMC Link